HRA GROUPS

TRUSTNET

AI-Powered Digital Identity & Fraud Intelligence Platform

TEAM ARJUNA

Chinni Anjaneyulu • G Swami Siva Rama Raju

AI • CYBERSECURITY • FULL STACK • MACHINE LEARNING • DATA INTELLIGENCE

“Don’t just verify identity. Continuously evaluate digital trust.”

01 THE CHALLENGE

Modern digital platforms process thousands of logins, account actions and transactions every minute. Traditional security systems often depend on isolated checks such as passwords, OTPs or fixed rules.

TrustNet challenges the team to build a system that evaluates digital trust by combining multiple signals and identifying suspicious patterns before they become serious security incidents.

02 PROBLEM STATEMENT

Build **TrustNet**, an intelligent digital identity and fraud-risk platform that analyzes identity information, device characteristics, session behavior, transaction patterns and historical activity to determine whether an interaction is legitimate, suspicious or potentially fraudulent.

The platform must provide both a **risk score** and an understandable explanation of the signals contributing to that score.

IDENTITY	BEHAVIOUR	SIGNALS	RISK	DECISION	ACTION

03 CORE PLATFORM CAPABILITIES

#	CAPABILITY	EXPECTATION
01	**Identity & Authentication**	Secure user accounts, authentication and role-based access.
02	**Behaviour Analysis**	Analyze login patterns, session activity and unusual user behaviour.
03	**Device Intelligence**	Track trusted and unfamiliar devices and identify abnormal device changes.
04	**Risk Scoring**	Calculate dynamic risk scores using multiple signals.
05	**Anomaly Detection**	Identify activity that deviates from historical or expected patterns.
06	**Fraud Alerts**	Generate real-time alerts and escalation workflows.
07	**Investigation Dashboard**	Give administrators a timeline and evidence view for suspicious activity.
08	**Explainable Decisions**	Show why a particular activity received its risk score.

04 SIGNATURE INNOVATION — BEHAVIOURAL RISK ENGINE

TrustNet should not treat every event independently. It should correlate multiple signals to detect patterns.

EXAMPLE SCENARIO

New device + unusual login time + sudden location change + unfamiliar transaction behaviour → TrustNet identifies a high-risk pattern → increases the risk score → explains the contributing signals → triggers an appropriate verification or review workflow.

05 ADVANCED CHALLENGE

- Graph-based relationships between users, devices, sessions and transactions
- Behavioural anomaly detection using machine-learning models
- Account takeover detection
- Synthetic or inconsistent identity detection
- Real-time event streaming and dynamic risk evaluation
- Adaptive authentication based on calculated risk
- AI investigation assistant that summarizes suspicious activity
- Predictive fraud-risk alerts before a high-risk action occurs
- Risk history, trends and attack-pattern analytics
- Explainable AI showing the strongest factors behind each decision

06 TECHNICAL EXPECTATIONS

- **Frontend:** Professional dashboards for users, security analysts and administrators.
- **Backend:** Secure APIs, authentication, event processing and investigation workflows.
- **Data Layer:** Store identities, devices, sessions, events, transactions and risk history.
- **AI/ML Layer:** Anomaly detection, scoring, classification and explainability.
- **Real-Time Layer:** Process security events and update risk status with minimal delay.
- **Security:** Apply secure design principles and protect sensitive identity information.

07 EXPECTED DELIVERABLES

- Working TrustNet application
- Secure authentication and identity workflow
- Real-time risk scoring and anomaly detection
- Administrator fraud-investigation dashboard
- Explainable risk decisions
- Demonstration with realistic suspicious and legitimate scenarios
- Architecture and AI/ML methodology documentation
- Deployment-ready prototype

08 EVALUATION FRAMEWORK

AREA	WEIGHT	WHAT MATTERS
Innovation	25%	Originality and strength of the trust-intelligence concept
Technical Depth	25%	Architecture, engineering quality and reliability
AI / ML Intelligence	25%	Detection quality, risk scoring and explainability
Security Design	15%	Secure architecture and responsible handling of data
UI / UX & Demo	10%	Clarity, usability and strength of the final demonstration

FINAL CHALLENGE	Build a platform that does not simply verify identity. It must continuously evaluate **digital trust** by connecting identity, behaviour, relationships and risk — then turn that intelligence into an actionable security decision.